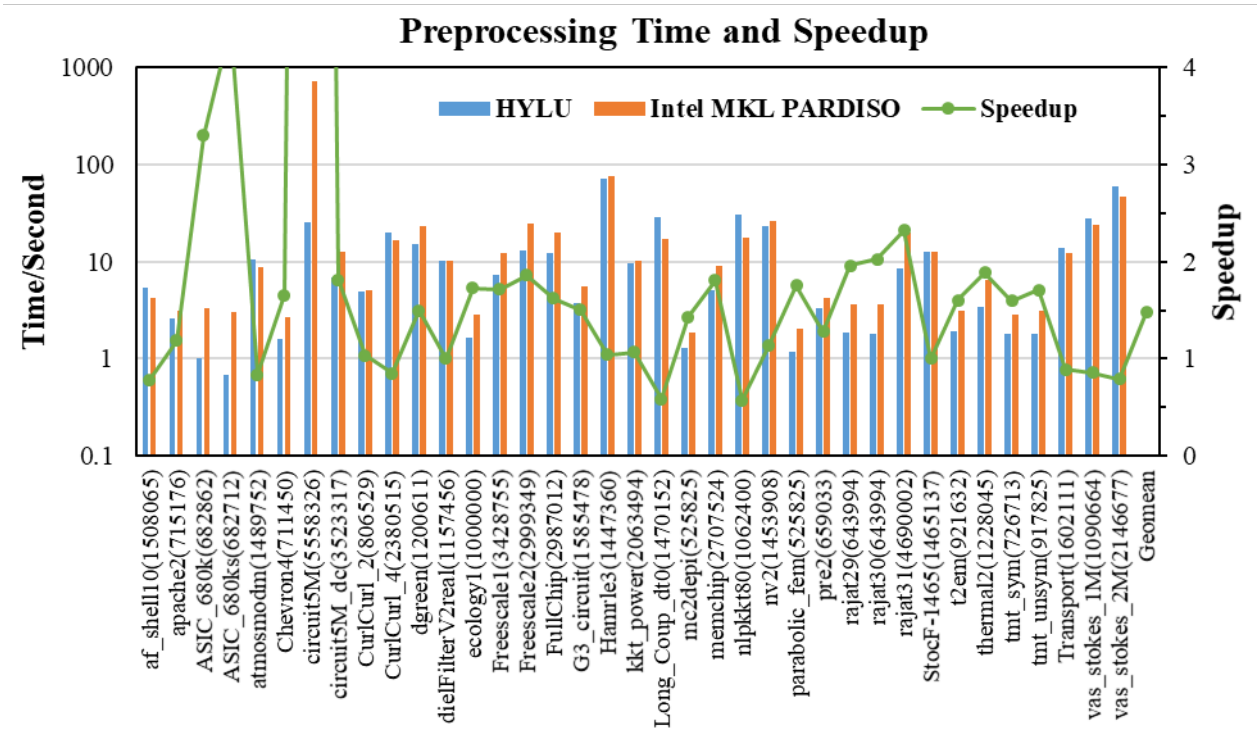
Test Results of HYLU and Performance Comparisons with Intel MKL PARDISO

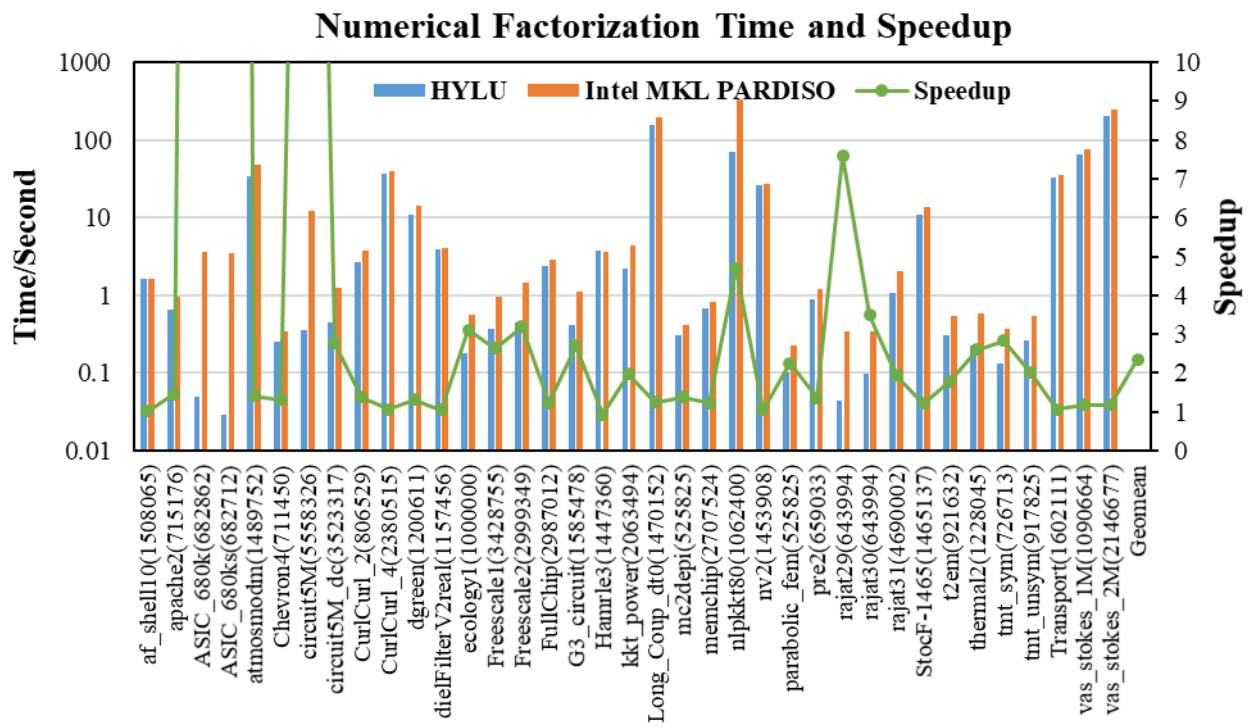
The experiments were carried out on a Linux server. The main hardware and software configurations are listed in the following table. All results presented in this document are wall-time measurements from **16-thread parallel execution**.

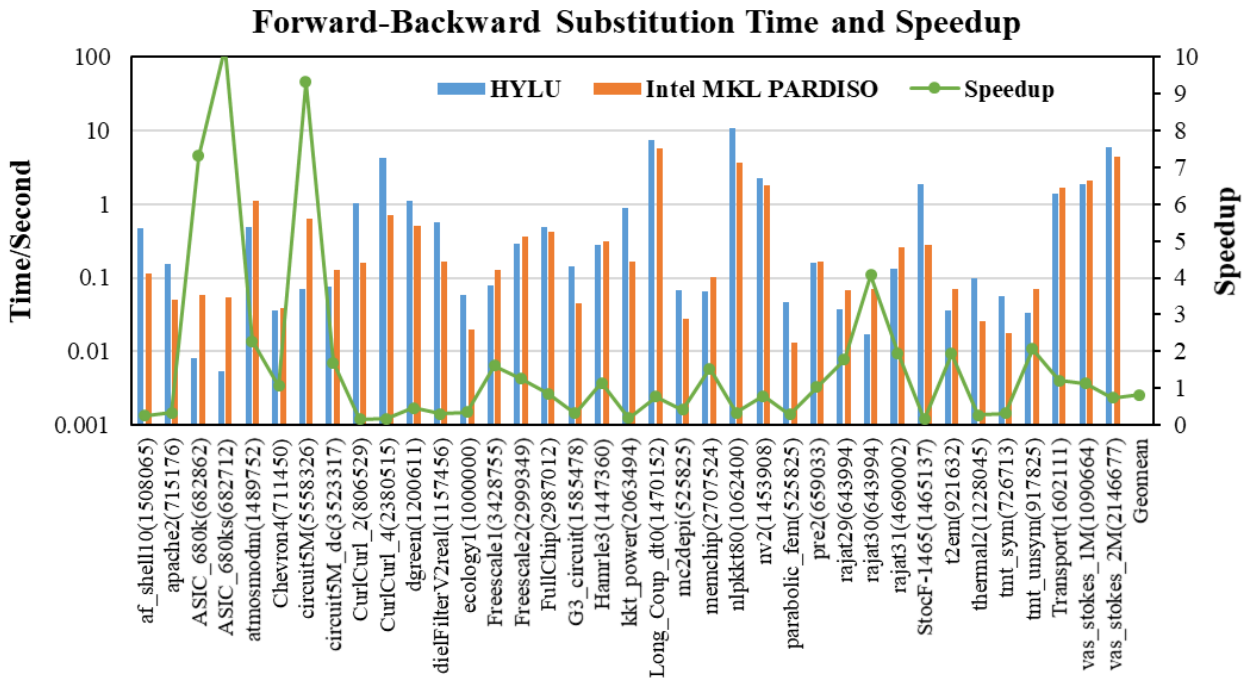
CPU	Intel Xeon Gold 6130 @ 2.1GHz
Memory	256GB
Operating system	Ubuntu 24.04 LTS for tests/Rocky Linux 8.10 for compiling HYLU library
Compiler	gcc 13.3.0 for compiling test code/gcc 8.5.0 for compiling HYLU library
MKL version	2025.3
HYLU version	20260331
Benchmarks	37 matrices from SuiteSparse Matrix Collection, dimensions from 525,825 to 5,558,326

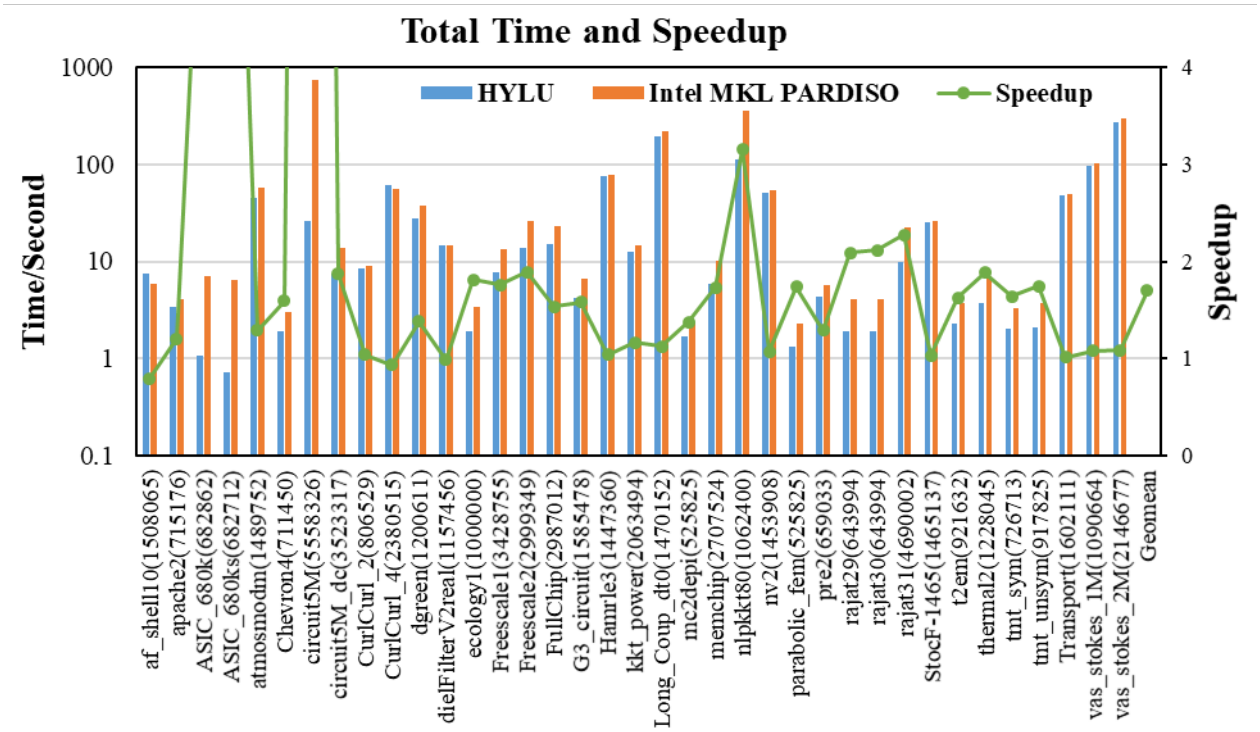
1. One-Time Solve

On geometric mean, HYLU achieves a **2.36X** speedup in numerical factorization compared with Intel MKL PARDISO, while the preprocessing phase is **1.48X** faster. The forward-backward substitution phase of HYLU is slightly slower and is 18% slower on geometric mean than that of Intel MKL PARDISO. For the total one-time solve time (preprocessing + numerical factorization + forward-backward substitution), HYLU is **1.70X** faster than Intel MKL PARDISO on geometric mean.




Numerical Factorization Time and Speedup
HYLU
Intel MKL PARDISO
Speedup
Time/Second
Speedup
1000
100
10
1
0.1
0.01
10
9
8
7
6
5
4
3
2
1
0
af_shell10(1508065)
apache2(715176)
ASIC_680k(682862)
ASIC_680ks(682712)
atmosmodm(1489752)
Chevron4(711450)
circuit5M(5558326)
circuit5M_dc(3523317)
CurlCurl_2(806529)
CurlCurl_4(2380515)
dgreen(1200611)
dielFilterV2real(1157456)
ecology1(1000000)
Freescale1(3428755)
Freescale2(2999349)
FullChip(2987012)
G3_circuit(1585478)
Hamrle3(1447360)
kkt_power(2063494)
Long_Coup_dt0(1470152)
mc2depi(525825)
memchip(2707524)
nlpkkt80(1062400)
nv2(1453908)
parabolic_fem(525825)
pre2(659033)
rajat29(643994)
rajat30(643994)
rajat31(4690002)
StocF-1465(1465137)
t2em(921632)
thermal2(1228045)
tmt_sym(726713)
tmt_unsym(917825)
Transport(1602111)
vas_stokes_1M(1090664)
vas_stokes_2M(2146677)
Geomean

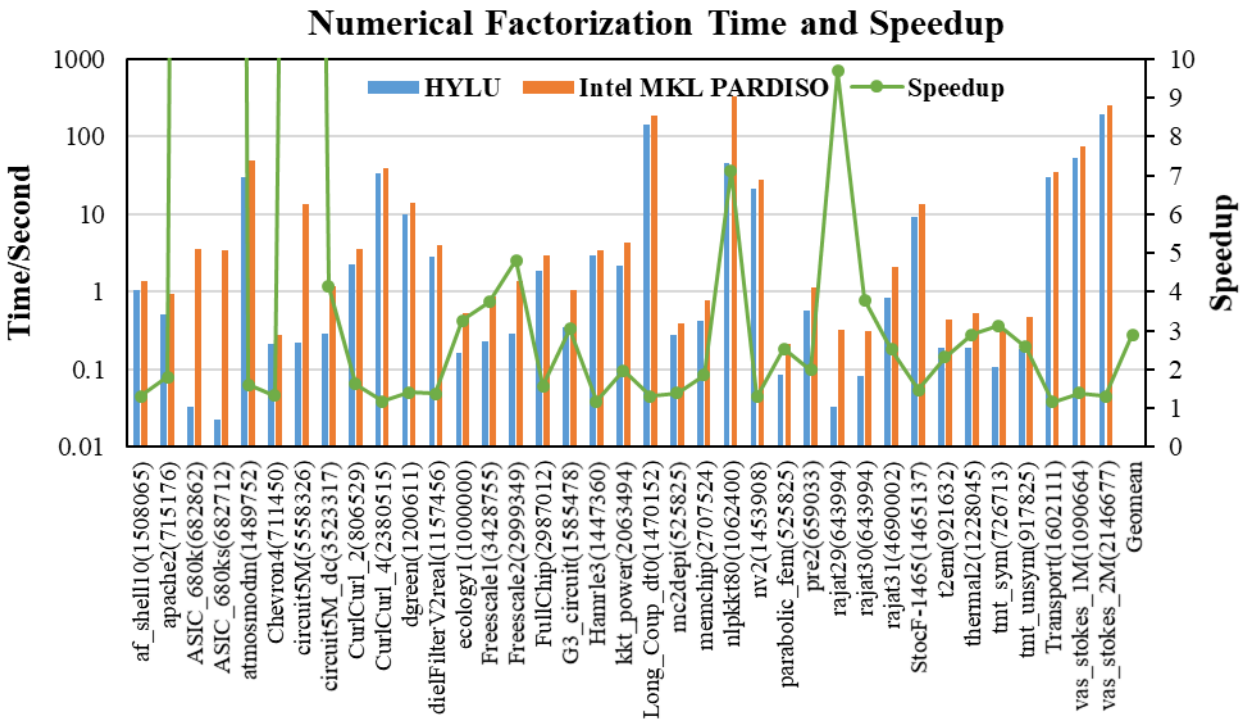

Forward-Backward Substitution Time and Speedup
HYLU
Intel MKL PARDISO
Speedup
Time/Second
Speedup
100
10
1
0.1
0.01
0.001
10
9
8
7
6
5
4
3
2
1
0
af_shell10(1508065)
apache2(715176)
ASIC_680k(682862)
ASIC_680ks(682712)
atmosmodm(1489752)
Chevron4(711450)
circuit5M(5558326)
circuit5M_dc(3523317)
CurlCurl_2(806529)
CurlCurl_4(2380515)
dgreen(1200611)
dielFilterV2real(1157456)
ecology1(1000000)
Freescale1(3428755)
Freescale2(2999349)
FullChip(2987012)
G3_circuit(1585478)
Hamrle3(1447360)
kkt_power(2063494)
Long_Coup_dt0(1470152)
mc2depi(525825)
memchip(2707524)
nlpkkt80(1062400)
nv2(1453908)
parabolic_fem(525825)
pre2(659033)
rajat29(643994)
rajat30(643994)
rajat31(4690002)
StocF-1465(1465137)
t2em(921632)
thermal2(1228045)
tmt_sym(726713)
tmt_unsym(917825)
Transport(1602111)
vas_stokes_1M(1090664)
vas_stokes_2M(2146677)
Geomean

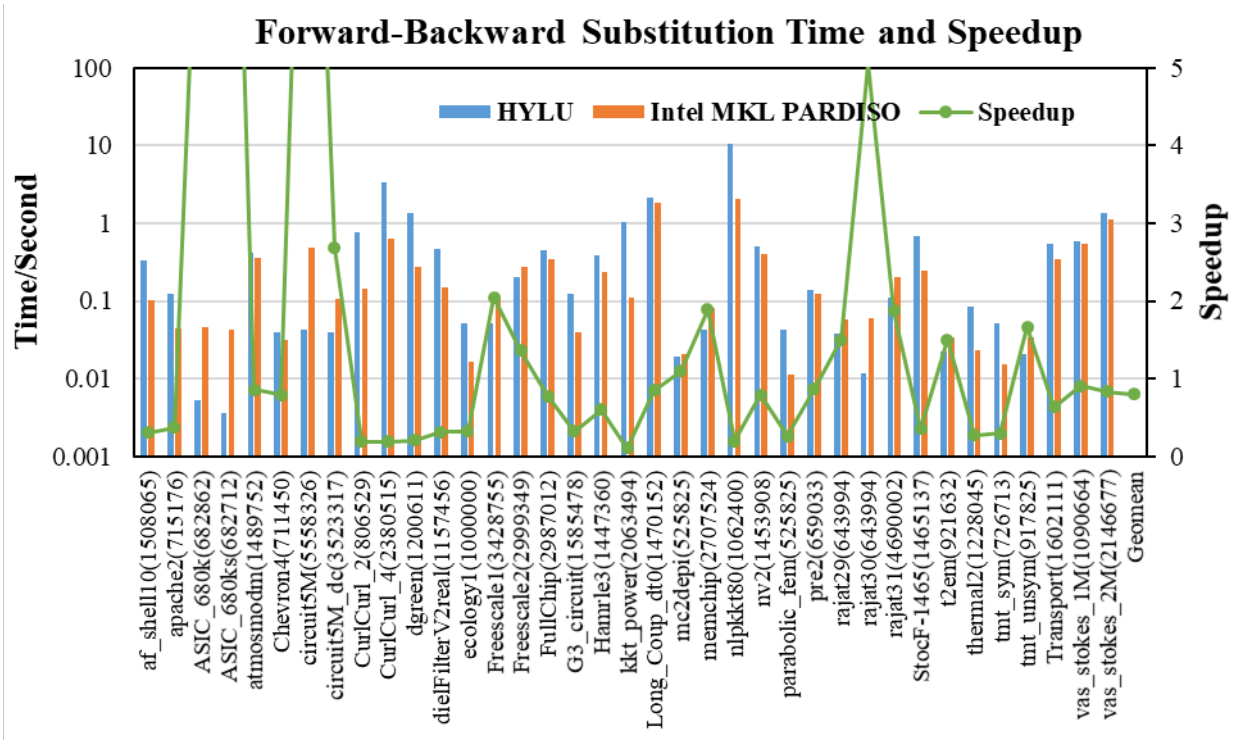


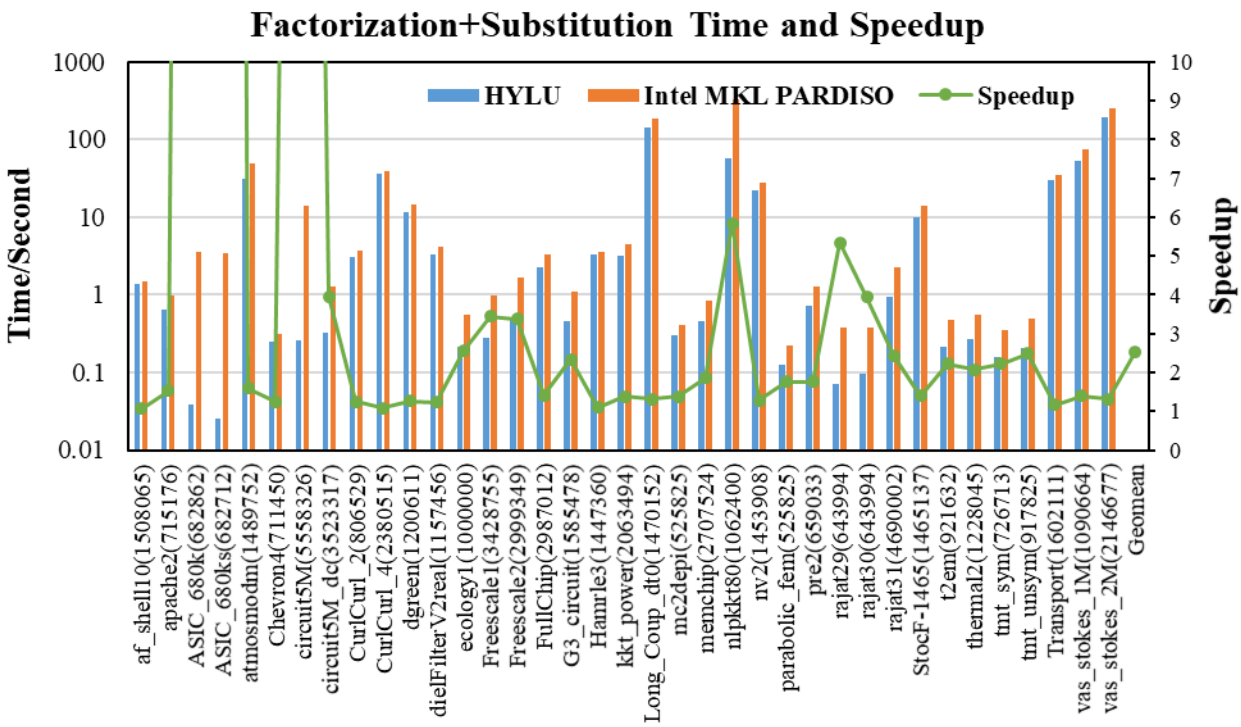
Another advantage of HYLU is its stable performance for various sparsities. Intel MKL PARDISO generates a huge quantity of fill-ins for some benchmarks (ASIC_680k, ASIC_680ks, circuit5M, and nlpkkt80), and the performance of these matrices is poor.

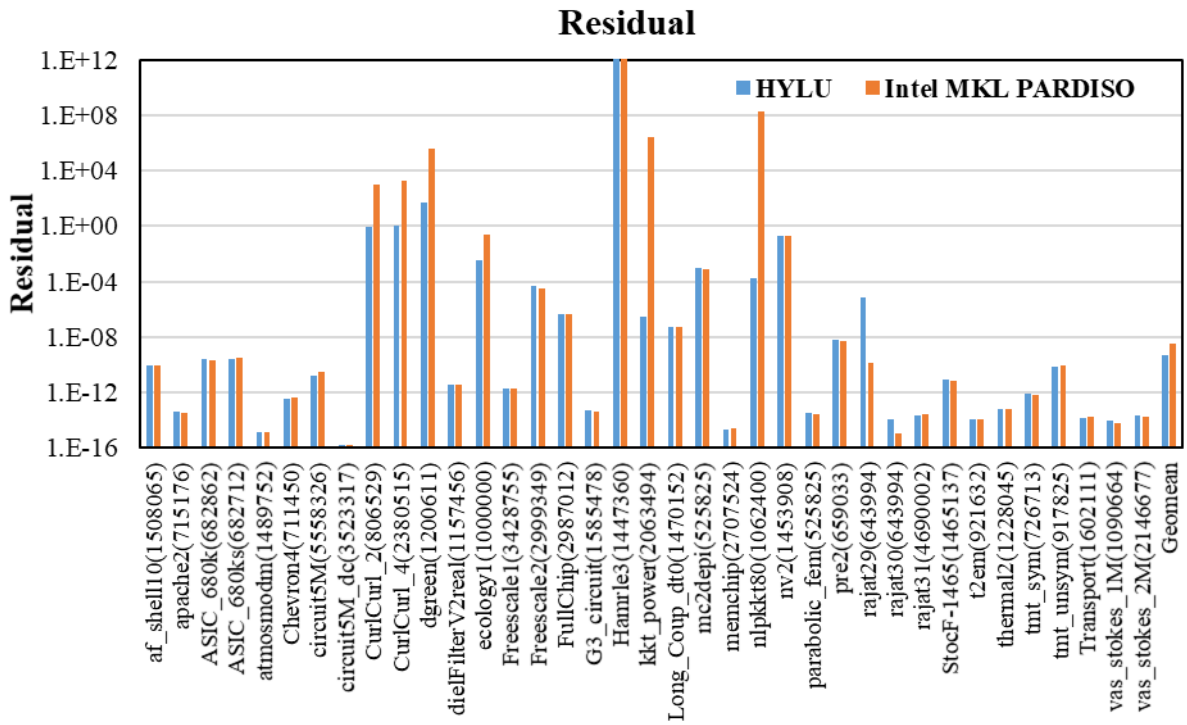
2. Repeated Solve

HYLU offers an optimization option for repeated solve of linear systems with an identical sparse pattern in the coefficient matrix. In this case, HYLU achieves a **2.90X** geometric mean speedup in numerical factorization over Intel MKL PARDISO, while the forward-backward substitution phase is slightly slower (20% slower on geometric mean). When comparing the total time of numerical factorization and forward-backward substitution, HYLU is faster than Intel MKL PARDISO for **ALL** of the tested benchmarks and the geometric mean of speedups is **2.53X**. In the repeated solve scenario, the preprocessing phase is on geometric mean 1.75X slower than that of the one-time solve scenario. However, in the repeated solve scenario, the pre-processing time is less important, as it is executed only once.


Numerical Factorization Time and Speedup
HYLU
Intel MKL PARDISO
Speedup
Time/Second
Speedup
1000
100
10
1
0.1
0.01
10
9
8
7
6
5
4
3
2
1
0
af_shell10(1508065)
apache2(715176)
ASIC_680k(682862)
ASIC_680ks(682712)
atmosmodm(1489752)
Chevron4(711450)
circuit5M(5558326)
circuit5M_dc(3523317)
CurlCurl_2(806529)
CurlCurl_4(2380515)
dgreen(1200611)
dielFilterV2real(1157456)
ecology1(1000000)
Freescale1(3428755)
Freescale2(2999349)
FullChip(2987012)
G3_circuit(1585478)
Hamrle3(1447360)
kkt_power(2063494)
Long_Coup_dt0(1470152)
mc2depi(525825)
memchip(2707524)
nlpkkt80(1062400)
nv2(1453908)
parabolic_fem(525825)
pre2(659033)
rajat29(643994)
rajat30(643994)
rajat31(4690002)
StocF-1465(1465137)
t2em(921632)
thermal2(1228045)
tmt_sym(726713)
tmt_unsym(917825)
Transport(1602111)
vas_stokes_1M(1090664)
vas_stokes_2M(2146677)
Geomean


Forward-Backward Substitution Time and Speedup
HYLU
Intel MKL PARDISO
Speedup
Time/Second
Speedup
100
10
1
0.1
0.01
0.001
5
4
3
2
1
0
af_shell10(1508065)
apache2(715176)
ASIC_680k(682862)
ASIC_680ks(682712)
atmosmodm(1489752)
Chevron4(711450)
circuit5M(5558326)
circuit5M_dc(3523317)
CurlCurl_2(806529)
CurlCurl_4(2380515)
dgreen(1200611)
dielFilterV2real(1157456)
ecology1(1000000)
Freescale1(3428755)
Freescale2(2999349)
FullChip(2987012)
G3_circuit(1585478)
Hamrle3(1447360)
kkt_power(2063494)
Long_Coup_dt0(1470152)
mc2depi(525825)
memchip(2707524)
nlpkkt80(1062400)
nv2(1453908)
parabolic_fem(525825)
pre2(659033)
rajat29(643994)
rajat30(643994)
rajat31(4690002)
StocF-1465(1465137)
t2em(921632)
thermal2(1228045)
tmt_sym(726713)
tmt_unsym(917825)
Transport(1602111)
vas_stokes_1M(1090664)
vas_stokes_2M(2146677)
Geomean



3. Accuracy

The residual is calculated as $\frac{\|\boldsymbol{Ax}-\boldsymbol{b}\|_1}{\|\boldsymbol{b}\|_1}$. On geometric mean, HYLU achieves **an order of magnitude higher accuracy** than Intel MKL PARDISO. This is mainly due to better control of pivoting and iterative refinement of HYLU, where the latter, however, introduces some overhead to the forward-backward substitution phase. Both HYLU and Intel MKL PARDISO fail to accurately solve Hamrle3, due to its extremely large condition number.